Example 4.2

In the Fig.4.35 $ABCD$ is a parallelogram, P and Q are the mid-points of sides AB and DC respectively. Show that $APCQ$ is a parallelogram.

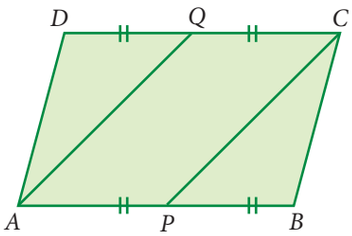


Fig. 4.35

Solution

Since P and Q are the mid points of

AB and DC respectively

Therefore $AP = \frac{1}{2} AB$ and

$QC = \frac{1}{2} DC$ (1)

But $AB = DC$ (Opposite sides of a parallelogram are equal)

$\Longrightarrow$ $\frac{1}{2} AB = \frac{1}{2} DC$

$\Longrightarrow$ $AP = QC$ (2)

Also , $AB \parallel DC$

$\Longrightarrow$ $AP \parallel QC$ (3) [$\because$ $ABCD$ is a parallelogram]

Thus , in quadrilateral $APCQ$ we have $AP = QC$ and $AP \parallel QC$ [from (2) and (3)]

Hence , quadrilateral $APCQ$ is a parallelogram.

Example 4.3

$ABCD$ is a parallelogram Fig.4.36 such that $\angle BAD = 120^{\circ}$ and AC bisects $\angle BAD$ show that $ABCD$ is a rhombus.

Solution

Given $\angle BAD = 120°$ and AC bisects $\angle BAD$

$\angle BAC = \frac{1}{2} \times 120° = 60°$

$\angle 1 = \angle 2 = 60°$

$AD \parallel BC$ and AC is the traversal

$\angle 2 = \angle 4 = 60°$

ΔABC is isosceles triangle [$\because \angle 1 = \angle 4 = 60°$]

$\Longrightarrow$ $AB = BC$

Parallelogram $ABCD$ is a rhombus.

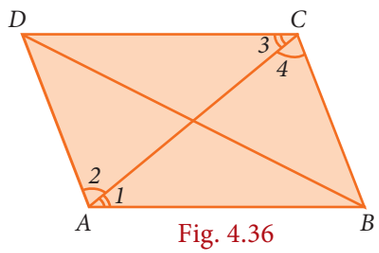


Fig. 4.36

4-Geometry_Term1.indd 156 26-12-2019 11:41:50